



Figure 2: Time to install SATA drivers



Figure 3: This is our XP Mode

In this screen, make sure **Boot Hard Disk (Primary Master)** is checked; then select **Create New Hard Disk** and click **Next** and again **Next**. Here, you should choose **Fixed Size Disk** for optimum performance, but a **Dynamically Allocated** one will do, too. A fixed-size disk will have all the hard disk space allocated at the beginning, while a dynamically allocated one will keep on expanding the disk size as the data grows in size. After making your selection, hit **Next**. Make sure the size is at least 10 GB, and then hit **Next** and finally **Finish**. Again, at the summary screen, hit **Finish**. We have just created the Virtual Machine. Now it's time to fine-tune it.

Make sure the Windows XP VM is selected, and hit the **Settings** button on the toolbar of VirtualBox. Move to the **System** section, and enable the checkbox for **IO APIC**. Move to the **Processor** tab and enable **PAE/NX** to make the **Execute Disable Bit** feature available for Windows. Also, if your CPU supports **AMD-V** or **Intel VT-x** (hardware-assisted virtualisation), you will be able to increase the number of virtual CPUs available for the use of the VM. It's a good idea to use half the physical number of cores—so on a dual core you would use one processor; on a quad core make it two. On my triple core, I'm using two processors since the host is Linux, but on a

Windows host this is not viable due to the performance penalty (the penalty, for some strange reason is Windows-only). Move to the **Acceleration** tab, and enable all the available features. The ones that are not available are greyed out. **VT-x** or **AMD-V** is the basic hardware-assist technology, and nested paging gives another huge performance boost.

Now move to the **Display** section and increase the **VRAM** to at least 64 MB. Also, enable **3D Acceleration**. Note that the **VRAM** is deducted from your graphics card memory (I'm endowed with 1 GB of it), so there's no penalty on available RAM. However, if all you have is an old Intel 865GV graphics card with 64 MB of VRAM, you'll have to make do with 12 MB of VRAM for the VM and no 3D acceleration. Once done, move on to the **Hard Disks** section.

Here, enable the checkbox for **Enable Additional Controller** and select **SATA (AHCI)**. In the attachments section, change the **HDD** slot from **IDE Primary Master** to **SATA Port 0**. Move to the **CD/DVD-ROM** section and enable the checkbox for **Mount CD/DVD Drive**. You may enable **Pass Through** if you want to write CDs in Windows, but at the moment, writing audio CDs is not possible and the host may behave in a bizarre manner.

Now move to the floppy section. Here comes the tricky part.

We enabled the SATA controller just now, as like the real thing, virtual SATA is a bit faster and less costly on the host to virtualise. However, Windows XP does not recognise any hard drives attached to the SATA controller, because it does not have the drivers. Therefore, Windows Setup exits with a 'No hard disks found' error. The solution is to use a floppy image loaded with the Intel Matrix Storage Technology SATA drivers (since the SATA controller is an Intel ICH8 chipset). Make sure the **unzip** command is installed, and then head to <http://tinyurl.com/matrix-drivers> to download the **f6flpy32.zip** file. Once it's downloaded, open a terminal and type in the following commands:

```
unzip f6flpy32.zip
unzip f6flpy32.exe
```

The EXE file is contained in the ZIP file and is another self-extracting ZIP file. Extracting it gives the file **F32.IMA**. This is a Windows Setup Driver Disk, which provides drivers to the initial Windows Setup to install Windows XP. Copy this file somewhere safe.

Now come back to VirtualBox. Check the **Mount Floppy Drive** checkbox, select **Image File** and click the folder icon beside it to launch the Virtual Media Manager. Hit **Add**, and then browse to the **F32.IMA** file (you'll have to change the **Files Of Type** to **All Files** to see the file). Hit **Select**. Now that this step is done, move to the **Shared Folders** section.

Here, you can specify a folder that will be accessible as a network share to Windows. This share can be mapped to a network drive, and you can use it as a storage drive. I find giving it access to the 'Public' directory in my home best serves this purpose. To do this, click the **Add Share** button (the blue folder with the **Plus** icon), and then on the **Folder Path** field browse to your 'Public' directory. The machine is now fully configured! It's time to install Windows XP.